Tier 2 | Knowledge and memory

Modules 07-09 | SDK 1.62 | 2026-10-29

Mental model

Retrieval finds evidence; generation writes from that evidence. Memory retains selected user information. Neither source text nor memory can grant tool authority.

RAG sequence

Load -> chunk -> embed/index -> retrieve -> rerank -> answer with citations. Compare retrieval scores and answer correctness separately.

Local setup

uv sync --group rag --group memory
uv run python modules/07_rag/examples/01_load_and_chunk.py --mock

Choose storage

BM25: lexical baseline. Vector search: semantic similarity. SQLite: explicit structured data. Managed Knowledge Bases and AgentCore Memory are optional AWS labs.

Evaluate evidence

Use held-out questions with expected sources. Measure missing evidence and incorrect citations. An answer that sounds fluent can still fail the case.

Memory boundaries

Ask for consent before saving preferences. Scope records by authenticated user. Treat recall as data. Test deletion and cross-user denial.

Sessions vs memory

A session restores conversation state. Long-term memory stores selected facts across sessions. Keep retention and deletion policies explicit.

Checkpoint and cleanup

Show a cited policy answer and consented preference recall. For optional AWS labs, destroy the resource stack and run scripts/find_leftovers.py --regions us-east-1.